

Assignment 1: Data Structure and Algorithms

M.M 50

Due Date: 27-08-2026

Note: There Are Four Sections. Each Section is compulsory.

Section A: Each Question Carry equal Marks (1 Marks Each)

- 1) Explain the difference between array traversal, insertion, and deletion. Give one example of each operation.
- 2) **Explain** the purpose of for, while, and do-while loops. Compare their execution conditions.
- 3) **Explain** the concept of class and object in C++. How are they related?
- 4) **Explain** the purpose of constructors and destructors in C++. Differentiate between their execution and purpose.
- 5) **Explain** the difference between public, private, and protected access specifiers with respect to accessibility.
- 6) **Explain** the FIFO principle of a queue. Differentiate between ENQUEUE and DEQUEUE operations.
- 7) What will be the output of following program code? Why?

```
#include <iostream.h>
using namespace std;
long int zip;    // Zip code
main( )
{
    zip = 021377;    // Use the zip code for Cambridge MA
    cout << "New York's zip code is: " << zip << '\n';
    return (0);
}
```
- 8) What is data structure. Explain with suitable examples.
- 9) What do you understand by Object oriented programming. How it is different from structural programming. Explain the features of Oop.

Section B: Each Question Carry equal Marks (2 Marks Each)

- 1) Apply linked-list concepts to create a singly linked list and perform insertion at the beginning, insertion at the end, deletion of a node, searching, and traversal operations.
- 2) **Apply** call by value and call by reference concepts to write a C++ program that swaps two numbers and demonstrates the difference between the two methods.
- 3) **Convert** the following infix expression into **postfix notation** and show the contents of the operator stack at each step: $((A+B)/C)*(D-E)$.
- 4) **Apply** the concept of array insertion and deletion to develop a C++ program that inserts an element at a specified position and deletes an element from a specified position.

- 5) **Apply** the LIFO principle to implement a stack using an array with PUSH, POP, PEEK, and DISPLAY operations. Handle overflow and underflow conditions.

Section C: Each Question Carry equal Marks (3 Marks Each)

- 1) **Analyze an array-based stack and identify the condition of the stack after the following operations. Explain where overflow or underflow can occur.**
PUSH(10)
PUSH(20)
POP()
POP()
POP()
PUSH(30)
PUSH(40)
- 2) Analyze the differences between an array, stack, queue, and singly linked list for insertion, deletion, memory allocation, and element access. Based on your analysis, determine which data structure is most suitable for implementing a browser back button, printer queue, and dynamic student record list, and justify your choices.
- 3) **Apply** the concept of classes and constructors to write a C++ program for a Student class. Use a **parameterized constructor** to initialize the student's name, roll number, and marks, and use a member function to display the details and calculate the grade.
- 4) **Analyze** the following postfix expression and develop a C++ program to evaluate it using a stack. Display the stack after processing every operator. $12\ 3\ 4\ *\ +\ 6\ 2\ /\ -$
- 5) Write a C++ program to implement linked list insertion.

Section D: Each Question Carry equal Marks (4 Marks Each)

- 1) Write a C++ program to calculate and the area of a triangle. Implement and display it using: Default Constructor, Parameterized constructor and Function overloading in one program.
- 2) Write a C++ program using data structure to sort a linked list by swapping data. The program must be general in nature and fully user friendly.
- 3) Write a C++ program using data structure to convert an Infix form to Postfix form. The program must be general in nature and fully user friendly.
- 4) Write a C++ program to implement a queue using an array. The program must be able to add and delete the data from the queue.